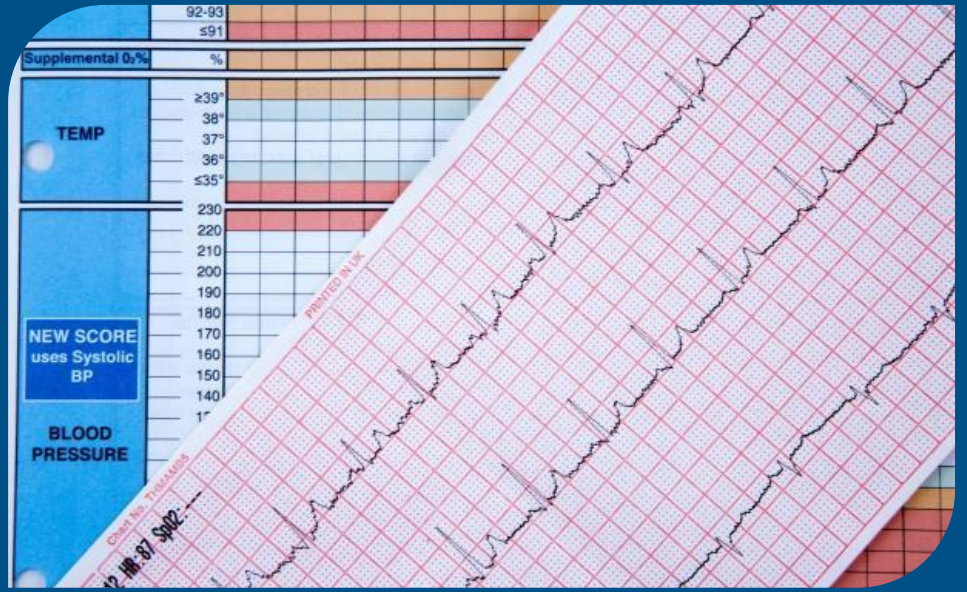
MIMIC SQL Tutorial
Heart Failure in MIMIC-IV: From Admission to Outcomes

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Introduction

- **Dataset:** MIMIC-IV v3.1
- **Platform:** Google BigQuery
- **Focus:** Heart-failure hospitalizations
- MIMIC-IV is a large de-identified electronic health record database containing hospital and ICU data, including patient demographics, diagnoses, admissions, medications, laboratory results, and ICU stays.
- For this SQL tutorial, heart-failure hospitalizations were identified using ICD-9 codes beginning with 428 and ICD-10 codes beginning with I50.
- The 10 queries follow the patient journey from diagnosis and demographics through hospitalization, ICU care, comorbidities, medications, laboratory findings, and outcomes.

- **Questions explored:** Who are the patients with heart failure? How do they enter and move through the hospital? What other conditions accompany heart failure? What treatments and abnormal laboratory findings are common? What are the mortality and 30-day readmission outcomes?



Which heart-failure diagnoses are most common?

```sql
-- Q1: Most common heart failure diagnoses in MIMIC-IV with percentage of all distinct HF admissions

WITH hf_codes AS (
    SELECT
        dx.hadm_id,
        dx.icd_version,
        dx.icd_code,
        UPPER(TRIM(dd.long_title)) AS diagnosis
    FROM `physionet-data.mimiciv_3_1_hosp.diagnoses_icd` AS dx
    INNER JOIN `physionet-data.mimiciv_3_1_hosp.d_icd_diagnoses` AS dd
        ON dx.icd_code = dd.icd_code
       AND dx.icd_version = dd.icd_version
    WHERE
          (dx.icd_version = 9 AND dx.icd_code LIKE '428%')
       OR (dx.icd_version = 10 AND dx.icd_code LIKE 'I50%')
),

total_hf AS (
    SELECT COUNT(DISTINCT hadm_id) AS total_hf_admissions
    FROM hf_codes
)

SELECT
    icd_version,
    icd_code,
    diagnosis,
    COUNT(DISTINCT hadm_id) AS hf_admissions,
    ROUND(
        100.0 * COUNT(DISTINCT hadm_id) /
        MAX(total_hf_admissions),
        1
    ) AS pct_of_hf_admissions
FROM hf_codes
CROSS JOIN total_hf
GROUP BY
    icd_version,
    icd_code,
    diagnosis
ORDER BY
    hf_admissions DESC;
```

This query identifies hospital admissions associated with heart failure using ICD-9 codes beginning with 428 and ICD-10 codes beginning with I50. It joins the diagnosis code table with the diagnosis description table, counts the number of distinct heart-failure admissions for each diagnosis, and calculates the percentage of all heart-failure admissions represented by each code.

Row	icd_version	icd_code	diagnosis	hf_admissions	pct_of_hf_admiss...
1	9	4280	CONGESTIVE HEART FAILURE, UNSPECIFIED	36606	45.4
2	10	I5032	CHRONIC DIASTOLIC (CONGESTIVE) HEART FAILURE	10787	13.4
3	9	42832	CHRONIC DIASTOLIC HEART FAILURE	8358	10.4
4	10	I5033	ACUTE ON CHRONIC DIASTOLIC (CONGESTIVE) HEART FAILURE	8118	10.1
5	10	I5022	CHRONIC SYSTOLIC (CONGESTIVE) HEART FAILURE	7507	9.3
6	9	42822	CHRONIC SYSTOLIC HEART FAILURE	6967	8.6
7	10	I5023	ACUTE ON CHRONIC SYSTOLIC (CONGESTIVE) HEART FAILURE	5685	7.1
8	9	42833	ACUTE ON CHRONIC DIASTOLIC HEART FAILURE	5506	6.8
9	9	42823	ACUTE ON CHRONIC SYSTOLIC HEART FAILURE	4476	5.6
10	10	I509	HEART FAILURE, UNSPECIFIED	3824	4.7

- The top 10 most commonheart failures codes are shown above (38 codes total).
- A hospitalization may include multiple HF diagnosis codes, so percentages could therefore overlap.
- Heart failure in MIMIC-IV is represented by a wide range of ICD-9 and ICD-10 diagnoses. The broad ICD-9 code for unspecified congestive heart failure is the most common, appearing in about 45% of heart-failure hospitalizations. More specific chronic and acute-on-chronic systolic and diastolic heart-failure diagnoses are also common, particularly among ICD-10 records.

What is the age and gender distribution of heart-failure patients?

```sql
-- Q2: Age and gender distribution at first heart failure hospitalization

WITH hf_admissions AS (
    SELECT DISTINCT
        subject_id,
        hadm_id
    FROM `physionet-data.mimiciv_3_1_hosp.diagnoses_icd`
    WHERE
          (icd_version = 9  AND icd_code LIKE '428%')
       OR (icd_version = 10 AND icd_code LIKE 'I50%')
),

hf_patient_admissions AS (
    SELECT
        hf.subject_id,
        hf.hadm_id,
        p.gender,
        a.age,
        a.admittime,
        ROW_NUMBER() OVER (
            PARTITION BY hf.subject_id
            ORDER BY a.admittime
        ) AS admission_number
    FROM hf_admissions AS hf
    INNER JOIN `physionet-data.mimiciv_3_1_derived.age` AS a
        ON hf.hadm_id = a.hadm_id
    INNER JOIN `physionet-data.mimiciv_3_1_hosp.patients` AS p
        ON hf.subject_id = p.subject_id
)

SELECT
    CASE
        WHEN age < 50 THEN '<50'
        WHEN age BETWEEN 50 AND 64 THEN '50-64'
        WHEN age BETWEEN 65 AND 79 THEN '65-79'
        ELSE '80+'
    END AS age_group,
    gender,
    COUNT(*) AS number_of_patients,
    ROUND(AVG(age), 1) AS mean_age
FROM hf_patient_admissions
WHERE admission_number = 1
GROUP BY
    age_group,
    gender
ORDER BY
    CASE age_group
        WHEN '<50' THEN 1
        WHEN '50-64' THEN 2
        WHEN '65-79' THEN 3
        WHEN '80+' THEN 4
    END,
    gender;
```

This query describes the age and gender distribution of patients at their first recorded heart-failure hospitalization. It uses a derived age table, assigns patients to age groups, and counts the number of male and female patients in each group while also calculating their mean age. Only the first heart-failure admission for each patient is included to avoid counting patients multiple times.

Row	age_group	gender	number_of_patie...	mean_age
1	<50	F	634	40.7
2	<50	M	1013	41.4
3	50-64	F	2238	58.5
4	50-64	M	3689	58.4
5	65-79	F	5233	72.6
6	65-79	M	6580	72.2
7	80+	F	6567	86.8
8	80+	M	5415	86.0

- Heart-failure patients in MIMIC-IV are predominantly older adults. About three-quarters of patients were age 65 or older at their first recorded heart-failure hospitalization. Men were more common in the younger age groups, while women were more numerous among patients age 80 and older.

How do heart-failure patients enter the hospital?

```sql
-- Q3: Admission type and admission location for heart failure hospitalizations

WITH hf_admissions AS (
    SELECT DISTINCT
        subject_id,
        hadm_id
    FROM `physionet-data.mimiciv_3_1_hosp.diagnoses_icd`
    WHERE
        (icd_version = 9  AND icd_code LIKE '428%')
      OR (icd_version = 10 AND icd_code LIKE 'I50%')
)

SELECT
    UPPER(TRIM(a.admission_type)) AS admission_type,
    UPPER(TRIM(a.admission_location)) AS admission_location,
    COUNT(DISTINCT a.hadm_id) AS number_of_admissions
FROM hf_admissions AS hf
INNER JOIN `physionet-data.mimiciv_3_1_hosp.admissions` AS a
    ON hf.hadm_id = a.hadm_id
GROUP BY
    admission_type,
    admission_location
ORDER BY
    number_of_admissions DESC;
```

This query examines how patients with heart failure enter the hospital. It groups heart-failure hospitalizations by admission type and admission location and counts the number of distinct admissions in each combination. The results show the most common pathways through which heart-failure patients are admitted to the hospital.

Row	admission_type	admission_location	number_of_admis...
1	EW EMER.	EMERGENCY ROOM	26185
2	URGENT	TRANSFER FROM HOSPITAL	7419
3	OBSERVATION ADMIT	PHYSICIAN REFERRAL	6425
4	OBSERVATION ADMIT	EMERGENCY ROOM	4409
5	EU OBSERVATION	EMERGENCY ROOM	3938
6	OBSERVATION ADMIT	WALK-IN/SELF REFERRAL	3493
7	SURGICAL SAME DAY ADMISSION	PHYSICIAN REFERRAL	2894
8	DIRECT EMER.	PHYSICIAN REFERRAL	2865
9	EW EMER.	PHYSICIAN REFERRAL	2702
10	EW EMER.	TRANSFER FROM HOSPITAL	2598
11	OBSERVATION ADMIT	TRANSFER FROM HOSPITAL	2598
12	DIRECT OBSERVATION	PHYSICIAN REFERRAL	2381
13	EW EMER.	WALK-IN/SELF REFERRAL	1983
14	ELECTIVE	PHYSICIAN REFERRAL	1688
15	EU OBSERVATION	WALK-IN/SELF REFERRAL	975

- The first 15 of 50 rows are shown above.
- Heart-failure hospitalizations most commonly entered through emergency pathways. About one-third of all HF admissions were emergency admissions originating in the emergency room. Transfers from other hospitals and several observation pathways were also common, while elective and surgical same-day admissions represented much smaller portions of the cohort.

How long do heart-failure hospitalizations last?

```sql
-- Q4: Length of stay for heart failure hospitalizations

WITH hf_admissions AS (
    SELECT DISTINCT
        subject_id,
        hadm_id
    FROM `physionet-data.mimiciv_3_1_hosp.diagnoses_icd`
    WHERE
        (icd_version = 9  AND icd_code LIKE '428%')
       OR (icd_version = 10 AND icd_code LIKE 'I50%')
),

hf_los AS (
    SELECT
        hf.subject_id,
        hf.hadm_id,
        TIMESTAMP_DIFF(
            a.dischtime,
            a.admittime,
            MINUTE
        ) / 1440.0 AS length_of_stay_days
    FROM hf_admissions AS hf
    INNER JOIN `physionet-data.mimiciv_3_1_hosp.admissions` AS a
        ON hf.hadm_id = a.hadm_id
    WHERE
        a.admittime IS NOT NULL
        AND a.dischtime IS NOT NULL
        AND a.dischtime >= a.admittime
)

SELECT
    CASE
        WHEN length_of_stay_days < 3 THEN 'Short (<3 days)'
        WHEN length_of_stay_days <= 7 THEN 'Medium (3–7 days)'
        ELSE 'Long (>7 days)'
    END AS los_group,

    COUNT(*) AS number_of_admissions,

    ROUND(
        100.0 * COUNT(*) / SUM(COUNT(*)) OVER (),
        1
    ) AS pct_of_admissions,

    ROUND(AVG(length_of_stay_days), 1) AS mean_los_days,
    ROUND(MIN(length_of_stay_days), 1) AS min_los_days,
    ROUND(MAX(length_of_stay_days), 1) AS max_los_days

FROM hf_los
GROUP BY los_group
ORDER BY
    CASE los_group
        WHEN 'Short (<3 days)' THEN 1
        WHEN 'Medium (3–7 days)' THEN 2
        WHEN 'Long (>7 days)' THEN 3
    END;
```

This query examines the length of hospital stay for heart-failure admissions. It calculates the number of days between admission and discharge, groups hospitalizations into short (less than 3 days), medium (3–7 days), and long (more than 7 days) stays, and reports the number and percentage of admissions in each group. It also calculates the average, minimum, and maximum length of stay for each category.

Row	los_group	number_of_admissions	pct_of_admissions	mean_los_days	min_los_days	max_los_days
1	Short (<3 days)	27007	33.5	1.7	0.0	3.0
2	Medium (3-7 days)	27524	34.2	4.8	3.0	7.0
3	Long (>7 days)	26059	32.3	14.9	7.0	296.0

- Admissions with discharge times earlier than admission times were excluded from the LOS analysis.
- Heart-failure hospitalizations were distributed fairly evenly across short, medium, and long stays. About 32% lasted more than seven days, and these long stays averaged approximately 15 days, suggesting that a substantial portion of heart-failure admissions involved prolonged inpatient care.

What percentage of heart-failure hospitalizations include ICU care?

```sql
-- Q5: Percentage of heart failure hospitalizations requiring ICU care

WITH hf_admissions AS (
    SELECT DISTINCT
        subject_id,
        hadm_id
    FROM `physionet-data.mimiciv_3_1_hosp.diagnoses_icd`
    WHERE
        (icd_version = 9  AND icd_code LIKE '428%')
      OR (icd_version = 10 AND icd_code LIKE 'I50%')
),

icu_admissions AS (
    SELECT DISTINCT
        hadm_id
    FROM `physionet-data.mimiciv_3_1_icu.icustays`
)

SELECT
    CASE
        WHEN icu.hadm_id IS NOT NULL THEN 'ICU'
        ELSE 'No ICU'
    END AS icu_status,

    COUNT(*) AS number_of_admissions,

    ROUND(
        100.0 * COUNT(*) / SUM(COUNT(*)) OVER (),
        1
    ) AS pct_of_hf_admissions

FROM hf_admissions AS hf

LEFT JOIN icu_admissions AS icu
    ON hf.hadm_id = icu.hadm_id

GROUP BY
    icu_status

ORDER BY
    number_of_admissions DESC;
```

This query determines how often heart-failure hospitalizations include an ICU stay. It identifies all heart-failure admissions and then uses a left join with the ICU-stay data to classify each hospitalization as either “ICU” or “No ICU.” The query reports the number and percentage of heart-failure hospitalizations in each group.

Row	icu_status	number_of_admissions	pct_of_hf_admissions
1	No ICU	57982	71.9
2	ICU	22629	28.1

- An important query design choice is to reduce icustays to one row per hospital admission before joining, because a single hospitalization can contain multiple ICU stays. Otherwise, the ICU group could be overcounted.
- Using a LEFT JOIN matters: an INNER JOIN would return only HF admissions that had an ICU stay and would make it impossible to measure the No ICU group.
- Most heart-failure hospitalizations did not require ICU care. About 28% of HF admissions included an ICU stay, while roughly 72% were managed outside the ICU.

Which ICU units care for heart-failure patients, and how long do they stay?

```sql
-- Q6: ICU care units and ICU length of stay for heart failure hospitalizations

WITH hf_admissions AS (
    SELECT DISTINCT
        subject_id,
        hadm_id
    FROM `physionet-data.mimiciv_3_1_hosp.diagnoses_icd`
    WHERE
        (icd_version = 9  AND icd_code LIKE '428%')
     OR (icd_version = 10 AND icd_code LIKE 'I50%')
)

SELECT
    UPPER(TRIM(icu.first_careunit)) AS first_icu_unit,

    COUNT(*) AS number_of_icu_stays,

    COUNT(DISTINCT icu.hadm_id) AS number_of_hospital_admissions,

    ROUND(AVG(icu.los), 1) AS mean_icu_los_days,

    ROUND(MIN(icu.los), 1) AS min_icu_los_days,

    ROUND(MAX(icu.los), 1) AS max_icu_los_days

FROM hf_admissions AS hf

INNER JOIN `physionet-data.mimiciv_3_1_icu.icustays` AS icu
    ON hf.hadm_id = icu.hadm_id

WHERE
    icu.los IS NOT NULL
    AND icu.los >= 0
    AND icu.first_careunit IS NOT NULL

GROUP BY
    first_icu_unit

ORDER BY
    number_of_icu_stays DESC;
```

This query examines heart-failure hospitalizations that included ICU care. It groups ICU stays by the first ICU care unit entered and reports the number of ICU stays, the number of distinct hospital admissions, and the average, minimum, and maximum ICU length of stay for each unit. This shows which ICU settings most commonly care for heart-failure patients and how long patients typically remain in those units.

Row	first_icu_unit	number_of_icu_stays	number_of_hospi...	mean_icu_los_days	min_icu_los_days	max_icu_los_days
1	MEDICAL INTENSIVE CARE UNIT (MICU)	6620	6088	4.1	0.0	140.1
2	CORONARY CARE UNIT (CCU)	6569	6116	3.7	0.0	74.0
3	CARDIAC VASCULAR INTENSIVE CARE UNIT (CVICU)	4508	4179	4.6	0.0	103.5
4	MEDICAL/SURGICAL INTENSIVE CARE UNIT (MICU/SICU)	3889	3496	3.6	0.0	127.9
5	SURGICAL INTENSIVE CARE UNIT (SICU)	1932	1822	4.0	0.0	77.7
6	TRAUMA SICU (TSICU)	1476	1362	3.9	0.0	112.1
7	NEURO INTERMEDIATE	552	525	6.2	0.2	62.2
8	NEURO SURGICAL INTENSIVE CARE UNIT (NEURO SICU)	304	298	4.8	0.0	50.2
9	NEURO STEPDOWN	111	107	5.5	0.2	59.7
10	PACU	30	27	5.0	0.0	37.6
11	SURGERY/VASCULAR/INTERMEDIATE	24	23	13.5	1.1	43.6
12	INTENSIVE CARE UNIT (ICU)	7	7	3.5	1.1	6.2
13	SURGERY/TRAUMA	5	5	12.6	3.8	17.0
14	MEDICINE	5	3	10.7	2.6	18.0

- A single hospitalization can include more than one ICU stay. The difference between the two columns is a measure of repeat ICU utilization within the same hospitalization.
- Heart-failure patients requiring ICU care were most commonly admitted to the MICU, CCU, and CVICU. Average ICU length of stay in the major units was generally about 4 days. The number of ICU stays was slightly higher than the number of hospital admissions, indicating that some patients had multiple ICU stays during the same hospitalization.

What comorbidities are most common in heart-failure hospitalizations?

```sql
-- Q7: Most common comorbidities among heart failure hospitalizations

WITH hf_admissions AS (
    SELECT DISTINCT hadm_id
    FROM `physionet-data.mimiciv_3_1_hosp.diagnoses_icd`
    WHERE
        (icd_version = 9  AND icd_code LIKE '428%')
      OR (icd_version = 10 AND icd_code LIKE 'I50%')
),

hf_charlson AS (
    SELECT c.*
    FROM `physionet-data.mimiciv_3_1_derived.charlson` AS c
    INNER JOIN hf_admissions AS hf
        ON c.hadm_id = hf.hadm_id
),

comorbidity_rows AS (
    SELECT
        hadm_id,
        condition.comorbidity,
        condition.present
    FROM hf_charlson,
    UNNEST([
        STRUCT('Renal disease' AS comorbidity, renal_disease AS present),
        STRUCT('Chronic pulmonary disease', chronic_pulmonary_disease),
        STRUCT('Diabetes',
            GREATEST(diabetes_without_cc, diabetes_with_cc)),
        STRUCT('Myocardial infarction', myocardial_infarct),
        STRUCT('Peripheral vascular disease', peripheral_vascular_disease),
        STRUCT('Cerebrovascular disease', cerebrovascular_disease),
        STRUCT('Dementia', dementia),
        STRUCT('Liver disease',
            GREATEST(mild_liver_disease, severe_liver_disease)),
        STRUCT('Malignancy',
            GREATEST(malignant_cancer, metastatic_solid_tumor)),
        STRUCT('Rheumatic disease', rheumatic_disease)
    ]) AS condition
)

SELECT
    comorbidity,
    SUM(present) AS number_of_admissions,
    ROUND(100.0 * AVG(present), 1) AS pct_of_hf_admissions
FROM comorbidity_rows
GROUP BY comorbidity
ORDER BY number_of_admissions DESC;
```

This query identifies major comorbid conditions that occur alongside heart failure. It joins heart-failure hospitalizations with the MIMIC-IV derived Charlson comorbidity data and calculates the number and percentage of heart-failure admissions associated with each condition. Heart failure itself is excluded because it defines the study cohort. The results show which additional chronic diseases are most common among patients hospitalized with heart failure.

Row	comorbidity	number_of_admissions	pct_of_hf_admissions
1	Diabetes	37881	47.0
2	Renal disease	37286	46.3
3	Chronic pulmonary disease	27790	34.5
4	Myocardial infarction	19926	24.7
5	Peripheral vascular disease	13569	16.8
6	Malignancy	9169	11.4
7	Cerebrovascular disease	7991	9.9
8	Liver disease	6503	8.1
9	Dementia	4288	5.3
10	Rheumatic disease	3818	4.7

- Instead of a long sequence of UNION ALL statements, we use an array of STRUCTs with UNNEST(). This allows us to convert the Charlson columns into rows which can be grouped and summed.
- Heart-failure hospitalizations were associated with a high burden of comorbid disease. Diabetes and renal disease were each present in nearly half of HF admissions, while chronic pulmonary disease and prior myocardial infarction were also common. These results highlight the multisystem complexity of patients hospitalized with heart failure.

Which medications are most commonly prescribed during heart-failure hospitalizations?

```sql
-- Q8: Most commonly prescribed medications during heart failure hospitalizations

WITH hf_admissions AS (
    SELECT DISTINCT
        hadm_id
    FROM `physionet-data.mimiciv_3_1_hosp.diagnoses_icd`
    WHERE
        (icd_version = 9  AND icd_code LIKE '428%')
       OR (icd_version = 10 AND icd_code LIKE 'I50%')
),

hf_prescriptions AS (
    SELECT
        p.hadm_id,
        UPPER(TRIM(p.drug)) AS drug
    FROM `physionet-data.mimiciv_3_1_hosp.prescriptions` AS p
    INNER JOIN hf_admissions AS hf
        ON p.hadm_id = hf.hadm_id
    WHERE
        p.drug IS NOT NULL
        AND TRIM(p.drug) != ''
)

SELECT
    drug,
    COUNT(DISTINCT hadm_id) AS number_of_admissions,
    ROUND(
        100.0 * COUNT(DISTINCT hadm_id)
        / (SELECT COUNT(*) FROM hf_admissions),
        1
    ) AS pct_of_hf_admissions
FROM hf_prescriptions
GROUP BY drug
ORDER BY number_of_admissions DESC
LIMIT 20;
```

This query identifies the medications most commonly prescribed during heart-failure hospitalizations. It joins the heart-failure cohort with the MIMIC-IV prescriptions table, standardizes medication names, and counts the number and percentage of distinct heart-failure admissions in which each medication was prescribed. The results show which medications are most frequently ordered for patients hospitalized with heart failure.

Row	drug	number_of_admi...	pct_of_hf_admi...
1	SODIUM CHLORIDE 0.9% FLUSH	74741	92.7
2	ACETAMINOPHEN	60634	75.2
3	HEPARIN	52490	65.1
4	SENNA	50461	62.6
5	FUROSEMIDE	50084	62.1
6	ASPIRIN	46744	58.0
7	DOCUSATE SODIUM	43299	53.7
8	INSULIN	39324	48.8
9	POTASSIUM CHLORIDE	39222	48.7
10	DEXTROSE 50%	35144	43.6
11	GLUCAGON	35129	43.6
12	METOPROLOL TARTRATE	33055	41.0
13	ATORVASTATIN	32556	40.4
14	MAGNESIUM SULFATE	32545	40.4
15	0.9% SODIUM CHLORIDE	32274	40.0
16	BAG	28830	35.8
17	5% DEXTROSE	28511	35.4
18	GLUCOSE GEL	27199	33.7
19	ISO-OSMOTIC DEXTROSE	26363	32.7
20	POLYETHYLENE GLYCOL	25368	31.5

- The most frequently prescribed items during heart-failure hospitalizations included both general inpatient medications and cardiovascular therapies. Furosemide was prescribed in about 62% of admissions, while aspirin, metoprolol, and atorvastatin were also common. Insulin was prescribed in nearly half of HF admissions, which is consistent with the high prevalence of diabetes observed in the comorbidity analysis, although medication use alone does not establish a diabetes diagnosis.

Which commonly performed blood laboratory tests are most often abnormal in heart-failure hospitalizations?

```sql
-- Q9: Common blood laboratory tests with abnormal results during heart failure
hospitalizations

WITH hf_admissions AS (
    SELECT DISTINCT hadm_id
    FROM `physionet-data.mimiciv_3_1_hosp.diagnoses_icd`
    WHERE
        (icd_version = 9  AND icd_code LIKE '428%')
     OR (icd_version = 10 AND icd_code LIKE 'I50%')
),

lab_by_admission AS (
    SELECT
        l.hadm_id,
        UPPER(TRIM(d.label)) AS lab_test,
        UPPER(TRIM(d.fluid)) AS fluid,

        MAX(
            CASE
                WHEN LOWER(l.flag) = 'abnormal' THEN 1
                ELSE 0
            END
        ) AS had_abnormal_result

    FROM `physionet-data.mimiciv_3_1_hosp.labevents` AS l

    INNER JOIN hf_admissions AS hf
        ON l.hadm_id = hf.hadm_id

    INNER JOIN `physionet-data.mimiciv_3_1_hosp.d_labitems` AS d
        ON l.itemid = d.itemid

    WHERE UPPER(TRIM(d.fluid)) = 'BLOOD'

    GROUP BY
        l.hadm_id,
        lab_test,
        fluid
)

SELECT
    lab_test,
    fluid,

    COUNT(*) AS admissions_tested,

    SUM(had_abnormal_result) AS admissions_with_abnormal_result,

    ROUND(
        100.0 * SUM(had_abnormal_result) / COUNT(*),
        1
    ) AS abnormal_admission_pct

FROM lab_by_admission

GROUP BY
    lab_test,
    fluid

-- Include only tests performed in at least 10,000 HF admissions
HAVING COUNT(*) >= 10000

ORDER BY
    abnormal_admission_pct DESC,
    admissions_tested DESC

LIMIT 20;
```

This query examines commonly performed blood laboratory tests during heart-failure hospitalizations. For each test, it counts the number of HF admissions in which the test was performed and determines how many of those admissions had at least one abnormal result. Only tests performed in at least 10,000 HF admissions are included, allowing the analysis to focus on commonly used laboratory tests rather than rare or highly specialized tests.

Row	lab_test	fluid	admissions_tested	admissions_with...	abnormal_admission_pct
1	PO2	BLOOD	24668	23717	96.1
2	GLUCOSE	BLOOD	75821	68597	90.5
3	RED BLOOD CELLS	BLOOD	75509	67918	89.9
4	HEMOGLOBIN	BLOOD	75592	67880	89.8
5	NTPROBNP	BLOOD	14880	13141	88.3
6	HEMATOCRIT	BLOOD	76212	66057	86.7
7	RDW-SD	BLOOD	41664	35007	84.0
8	UREA NITROGEN	BLOOD	76681	61560	80.3
9	INR(PT)	BLOOD	61590	48787	79.2
10	PCO2	BLOOD	24663	19491	79.0
11	PT	BLOOD	61586	48239	78.3
12	FREE CALCIUM	BLOOD	13757	10459	76.0
13	LYMPHOCYTES	BLOOD	34321	26054	75.9
14	NEUTROPHILS	BLOOD	34321	25316	73.8
15	PH	BLOOD	25525	18278	71.6
16	VANCOMYCIN	BLOOD	11120	7349	66.1
17	CREATININE	BLOOD	76746	50274	65.5
18	ABSOLUTE NEUTROPHIL COUNT	BLOOD	20537	13421	65.4
19	TROPONIN T	BLOOD	39040	25045	64.2
20	ABSOLUTE LYMPHOCYTE COUNT	BLOOD	20624	12866	62.4

- Common blood tests showed a high frequency of abnormal results during heart-failure hospitalizations. NT-proBNP was abnormal in about 88% of admissions in which it was measured, while creatinine was abnormal in about 66%, consistent with the high burden of renal disease seen earlier. Abnormal glucose, blood-count, coagulation, and blood-gas results were also common, indicating substantial multisystem illness among hospitalized heart-failure patients.

What are the in-hospital mortality and 30-day readmission rates for heart-failure hospitalizations?

```sql
-- Q10: In-hospital mortality and 30-day readmission among heart failure hospitalizations

WITH hf_admissions AS (
    SELECT DISTINCT hadm_id
    FROM `physionet-data.mimiciv_3_1_hosp.diagnoses_icd`
    WHERE
        (icd_version = 9  AND icd_code LIKE '428%')
        OR (icd_version = 10 AND icd_code LIKE 'I50%')
),

-- Calculate each patient's next hospital admission using the complete admissions timeline
all_admissions AS (
    SELECT
        subject_id,
        hadm_id,
        admittime,
        dischtime,
        hospital_expire_flag,

        LEAD(admittime) OVER (
            PARTITION BY subject_id
            ORDER BY admittime, hadm_id
        ) AS next_admittime

    FROM `physionet-data.mimiciv_3_1_hosp.admissions`
),

-- Restrict to valid heart-failure index admissions and determine whether a surviving patient was
-- readmitted within 30 days
hf_outcomes AS (
    SELECT
        a.hadm_id,
        a.hospital_expire_flag,

        CASE
            WHEN a.hospital_expire_flag = 0
              AND a.next_admittime IS NOT NULL
              AND a.next_admittime > a.dischtime
              AND TIMESTAMP_DIFF(
                    a.next_admittime,
                    a.dischtime,
                    HOUR
                  ) <= 30 * 24
            THEN 1
            ELSE 0
        END AS readmitted_30_days

    FROM all_admissions AS a

    INNER JOIN hf_admissions AS hf
        ON a.hadm_id = hf.hadm_id

    WHERE
        a.admittime IS NOT NULL
        AND a.dischtime IS NOT NULL
        AND a.dischtime >= a.admittime
)

SELECT
    COUNT(*) AS total_hf_admissions,

    SUM(hospital_expire_flag) AS in_hospital_deaths,

    ROUND(
        100.0 * SAFE_DIVIDE(
            SUM(hospital_expire_flag),
            COUNT(*)
        ),
        1
    ) AS in_hospital_mortality_pct,

    COUNTIF(hospital_expire_flag = 0) AS survived_to_discharge,

    SUM(readmitted_30_days) AS readmitted_within_30_days,

    ROUND(
        100.0 * SAFE_DIVIDE(
            SUM(readmitted_30_days),
            COUNTIF(hospital_expire_flag = 0)
        ),
        1
    ) AS readmission_30_day_pct

FROM hf_outcomes;
```

This query examines two important outcomes of heart-failure hospitalizations: in-hospital mortality and 30-day readmission. It uses a window function to identify each patient's next hospital admission and determines whether patients who survived their heart-failure hospitalization returned to the hospital within 30 days of discharge. The query reports the in-hospital mortality rate for all heart-failure admissions and the 30-day readmission rate among patients who survived to discharge.

Row	total_hf_admissions	in_hospital_deaths	in_hospital_mortality_pct	survived_to_disc...	readmitted_within_30_da...	readmission_30_day_pct
1	80590	3997	5.0	76593	18925	24.7

- LEAD(admittime) allows us to fetch the admittime value from the next row immediately following the current row. If there is no next row (e.g., this was the patient's most recent hospitalization), LEAD() returns NULL.
- PARTITION BY subject_id groups the timeline independently for each unique patient. The calculation isolates Patient A's admissions from Patient B's admissions so that another patient's admission date is never fetched.
- Heart-failure hospitalizations had an in-hospital mortality rate of 5.0%. Among patients who survived to discharge, 24.7% were readmitted within 30 days, meaning roughly one in four surviving hospitalizations was followed by another admission within a month. This indicates a substantial risk of early rehospitalization after heart-failure discharge.

Bonus: The Story of Heart Failure Across MIMIC-IV, From Admissions to Outcomes

- Across approximately 80,600 heart-failure hospitalizations, the data show a population that is predominantly older, medically complex, and at meaningful risk for prolonged hospitalization and early readmission.
- **Who are the patients?** About three-quarters of patients were age 65 or older at their first recorded heart-failure hospitalization. The diagnosis data included both systolic and diastolic heart failure, with unspecified congestive heart failure being the most common individual code.
- **How are they hospitalized?** Emergency care was the most common route into the hospital, with emergency-room admissions accounting for about one-third of HF hospitalizations. Length of stay was broadly distributed: roughly one-third of admissions were short (<3 days), one-third medium (3–7 days), and one-third longer than 7 days.
- **How complex is their care?** About 28% of HF hospitalizations included ICU care, most commonly in the MICU, CCU, and CVICU. Comorbidities were frequent, especially diabetes (47%), renal disease (46%), chronic pulmonary disease (35%), and prior myocardial infarction (25%). Medication and laboratory results reinforced this picture of complex multisystem illness.
- **What are the outcomes?** In-hospital mortality was 5.0%. Among patients who survived to discharge, 24.7% were readmitted within 30 days, which is approximately one in four surviving hospitalizations.
- **Overall takeaway:** The 10 queries show that heart failure in MIMIC-IV is not simply a cardiac diagnosis. It commonly occurs in older patients with multiple chronic conditions, often requires substantial hospital resources, and is associated with a high rate of early rehospitalization after discharge.